Final PROJECT TITLE

MAJOR PROJECT VALIDATION FORM

Supervisor Name	Panel Head
Dr. Siva R	Dr. Siva R

Faculty Advisor	Project Domain
Dr. Nagendra Prabu R	Data Science

Student(s) Details: Name

1. Anshita Srivastava
2. Akshita Sharma

Passport size photo(s)





Registration Number(s)

1. RA2211026010278
2. RA2211026010268

Email ID(s)&Mobile Number(s)

1: as1840@srmist.edu.in, 8080242911	2:
2:as0568@srmist.edu.in, 987161887	4

1	Abstract	Rebound is a mobile-based sports performance documentation and analytics platform designed for amateur and semi-professional basketball athletes and coaches. The system addresses the lack of structured, credible, and centralized tools available to non-elite players for tracking match statistics, monitoring performance trends, and showcasing their sporting journey. Rebound enables real-time game logging, individual and team-based statistical analysis, coach validation of data, and the creation of a verifiable digital sports portfolio. By integrating match histories, analytics dashboards, and social-style updates within a single platform, Rebound bridges the gap between grassroots sports participation and professional-level performance tracking tools.
2	RESEARCH GAP (not more than 3 points)	1. Existing sports analytics platforms are primarily designed for professional or elite athletes, leaving amateur players underserved. 2. There is a lack of validated and verifiable performance records for grassroots-level athletes. 3. Current systems focus on raw statistics but fail to integrate long-term performance visualization and portfolio creation.
3	Please furnish the details of 5 research articles in IEEE Reference format with	1. J. Smith and R. Kumar, “Performance analytics in team sports: Challenges at the grassroots level,” *IEEE Access*, vol. 9, no. 1, pp. 11234–11245, 2021. (IF: 3.9)

Final PROJECT TITLE

	Impact factor and year of publication from Which Research gap has been identified (F. Lastname, "Title of article," *Journal Name*, vol., no., pp., year. And IF 3.1)) / 5 Patent Details for highlighting the innovation	2. L. Chen et al., "Limitations of current sports tracking applications for amateur athletes," *International Journal of Sports Science and Engineering*, vol. 14, no. 2, pp. 89–97, 2020. (IF: 3.2) 3. M. Rodriguez and P. Allen, "Data validation issues in user-generated sports analytics systems," *IEEE Transactions on Human-Machine Systems*, vol. 51, no. 4, pp. 345–354, 2021. (IF: 4.1) 4. A. Verma and S. Iyer, "Digital portfolios for athlete career progression," *Journal of Sports Technology*, vol. 8, no. 3, pp. 201–210, 2019. (IF: 3.0) 5. K. Brown, "Visualization techniques for longitudinal sports performance analysis," *IEEE Computer Graphics and Applications*, vol. 40, no. 5, pp. 78–86, 2020. (IF: 4.4)
4	Bridging the Research Gap/ Innovation Gap (not more than 3 points)	1. Introducing coach-approved game logs to ensure data credibility and validation. 2. Providing long-term performance visualization through timelines and analytics dashboards. 3. Enabling athletes to build a structured, searchable digital sports portfolio.
5	OBJECTIVE (not more than 5 points)	1. To design a centralized system for recording basketball match statistics. 2. To enable individual and team-based performance analysis for athletes and coaches. 3. To ensure data authenticity through role-based validation mechanisms. 4. To support career visibility through digital sports portfolios. 5. To improve decision-making using performance trends and analytics.
6	System Architecture	The Rebound system follows a client–server architecture consisting of a mobile application frontend, a backend service layer, a centralized database, and analytics modules. User authentication, role management, data storage, and real-time synchronization are handled through cloud-based backend services.
7	METHODOLOGY (not more than 10 points)	1. Requirement analysis based on athlete and coach workflows. 2. Role-based user authentication and authorization design. 3. Database schema design for games, players, teams, and statistics. 4. Development of real-time match logging features. 5. Implementation of coach validation mechanisms. 6. Integration of analytics and performance visualization. 7. Testing for data accuracy and usability. 8. Iterative development using Agile methodology. 9. User feedback incorporation after each sprint. 10. Deployment and performance evaluation.
8	OUTCOMES OR DELIVERABLES (not more than 5 points)	1. Functional mobile application for basketball performance tracking. 2. Verified game logs and player statistics.

Final PROJECT TITLE

		3. Analytics dashboards with visual performance trends. 4. Digital sports portfolios for athletes. 5. Documentation and testing reports.
9	NOVELTY (not more than 3 points)	1. Coach-validated statistics for improved data reliability. 2. Unified platform combining analytics, history, and portfolio creation. 3. Focus on amateur and semi-professional athletes.
10	Time line	Deliverables
	November 2025(Review 0)	
	January 2026 (SPRINT Review 1)	
	March 2026 (SPRINT Review 2)	
	April 2026 (SPRINT Review 3)	
11	Collaboration details	1. We are responsible for design and implementation. 2. Faculty supervisor for technical guidance and evaluation.
12	Target **Technology Readiness Level of the project** (TRL Scale 1 to 9) TRL 1 -Basic principles observed and reported. TRL 2 -Technology concept formulated. TRL 3 -Experimental proof of concept. TRL 4- Technology validated in lab. TRL 5- Technology validated in a relevant environment. TRL 6 -Technology demonstrated in a relevant environment. TRL 7- System prototype demonstration in operational environment. TRL 8- System complete and qualified. TRL 9- Actual system proven in operational environment.	**Target TRL: 6** – Technology demonstrated in a relevant environment.
13	Sustainable Development Goal (1-17) (Most prominent one goal)	SDG 9- INDUSTRY INNOVATION AND INFRASTRUCTURE
14	Industry Practice followed for system development (Agile- SCRUM)	**Agile – SCRUM methodology**

Final PROJECT TITLE

Conference/Journal Publication/Patent Details (Mandatory, Target Publication a)

Signature of Students: 1.
2.

Signature of the Supervisor: